

Uday Kiran Padhy

📞 9381569588 | ✉ kiranu941@gmail.com | 🌐 udaykp.dev | 🔗 udaykiranpadhy | 🐙 UdayKiranPadhy

Summary

Senior Software Engineer with 3+ years shipping production AI at Gainsight: LLM agents on LangGraph, RAG pipelines, and a from-scratch RL training stack for cloud agents. Comfortable owning fine-tuning, evaluation, and the FastAPI services that serve the models.

Experience

Gainsight — Hyderabad, India — August 2022 -- Present

Promoted twice: Associate Software Engineer → Software Engineer → Senior Software Engineer.

Senior Software Engineer — Aug 2025 -- Present

- Built customer-facing LLM features on Retrieval-Augmented Generation pipelines that ground model responses in tenant data, reducing hallucinations on customer-data Q&A.
- Shipped AI capabilities into Fortune 500 deployments (Intuit, Adobe) by aligning RAG quality, latency, and SLA targets with enterprise readiness.
- Externalized AI-serving APIs as public endpoints with authn/authz and scoped permissions; remediated critical Python CVEs flagged by Snyk across AI dependencies.

Software Engineer — Aug 2023 -- Aug 2025

- Architected multi-step LangGraph agents orchestrating retrieval, tool-use, and conditional reasoning to automate business decisioning previously handled by hand-written rules.
- Built FastAPI services hosting model inference, retrieval, and a user segmentation pipeline running across millions of users for targeted product surfaces.
- Scaled backend APIs serving AI traffic to seven-figure peak RPS at sub-50ms P99 latency via horizontal scaling and distributed caching; owned observability (metrics, logs, distributed tracing) at 99.9% uptime.

Associate Software Engineer — Aug 2022 -- Aug 2023

- Built unit, integration, and end-to-end test suites and Docker-based Python dev environments -- the testability foundation later inherited by AI services.

Projects

AWS RL Environment — GRPO, Unsloth, TRL, PyTorch, HuggingFace, FastAPI, OpenEnv

- End-to-end RL training pipeline for cloud-infrastructure agents: SFT then GRPO on Qwen2.5-Coder-3B over 120+ AWS tasks; SFT lifted exact-match accuracy from 39% to 89% on the eval set, GRPO added another +6 pp on the intermediate tier. Published the dataset, SFT and GRPO adapters, and a live HuggingFace Space.

Aakar — LLM Architecture Visualizer — PyTorch, HuggingFace Transformers, FastAPI, React, TypeScript, React Flow

- Interactive LLM-architecture explorer at aakar.udaykp.dev: introspects any HuggingFace model on PyTorch's meta device (no weights downloaded) and renders the real nn.Module tree as a zoomable diagram with fake-tensor forward-pass op traces, parameter/compute breakdowns, and side-by-side model comparison; any transformers-native family renders with zero per-family code.

LeetCode Companion — Gemini AI, FastAPI, React, Server-Sent Events, crawl4ai, Docker

- AI tutor at leetcode.udaykp.dev: streams Gemini-agent reasoning, formal complexity proofs (KaTeX), and multi-language annotated solutions over Server-Sent Events; FastAPI + crawl4ai problem extraction, React + TypeScript frontend, one-command Docker Compose deploy.

Technical Skills

AI & ML: RAG, LLM Agents, LangGraph, Fine-tuning (SFT, GRPO, RLHF), Vector Databases, Evaluation

Frameworks & Tooling: PyTorch, HuggingFace, TRL, Unsloth, OpenEnv, Gemini AI, crawl4ai

Languages: Python, TypeScript, SQL

Backend & Serving: FastAPI, React, Server-Sent Events, Docker, Microservices

Cloud & Infrastructure: AWS, GCP, Terraform, CI/CD

Education

Sagi Rama Krishnam Raju Engineering College — Bhimavaram, Andhra Pradesh — June 2018 -- July 2022

Bachelor of Technology in Electronics and Communication

Achievements

Smart India Hackathon: 6th place nationally; led a 4-person team out of 100+ teams to ship an accessibility application for physically challenged users -- eye-tracking via OpenCV plus speech-to-text and text-to-speech for hands-free computer control.

LeetCode Weekly Contest 393: Rank 233 worldwide out of ~25,000 participants; profile leetcode.com/u/lamUday.